

Name: Arnav Krishnan

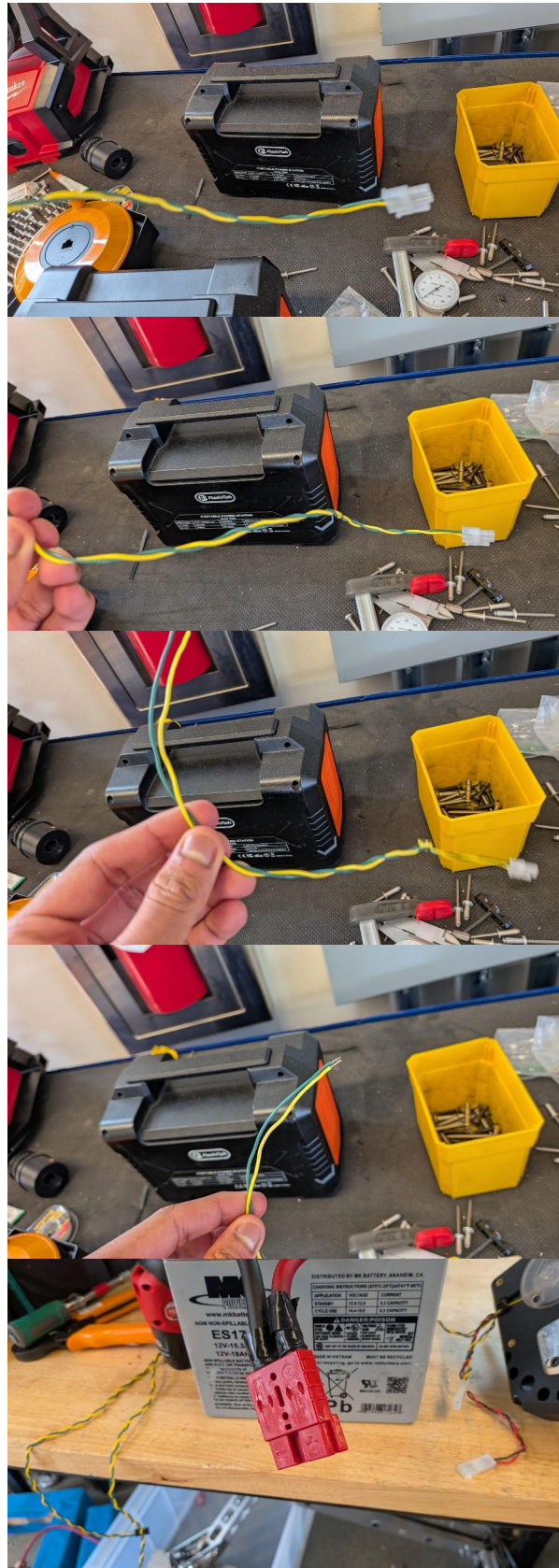
Report Date: Week of Sept 28-Oct 2

Work Completion Conflicts: Our lead wasn't here on Friday, or for the majority of Monday.

Note: I have the same amount of text as normal, it's just that I have more images.

Completed Tasks

Task 1, Wiring and Power, 1 period.



Starting the electrical rotation, we started mainly concerned with how power travels through the robot. Power starts at the 12V battery, passes through the main breaker (which is the robot's main on/off switch), and enters the power distribution hub ("PDH"). The hub splits it into channels, with each channel having its own fuse.

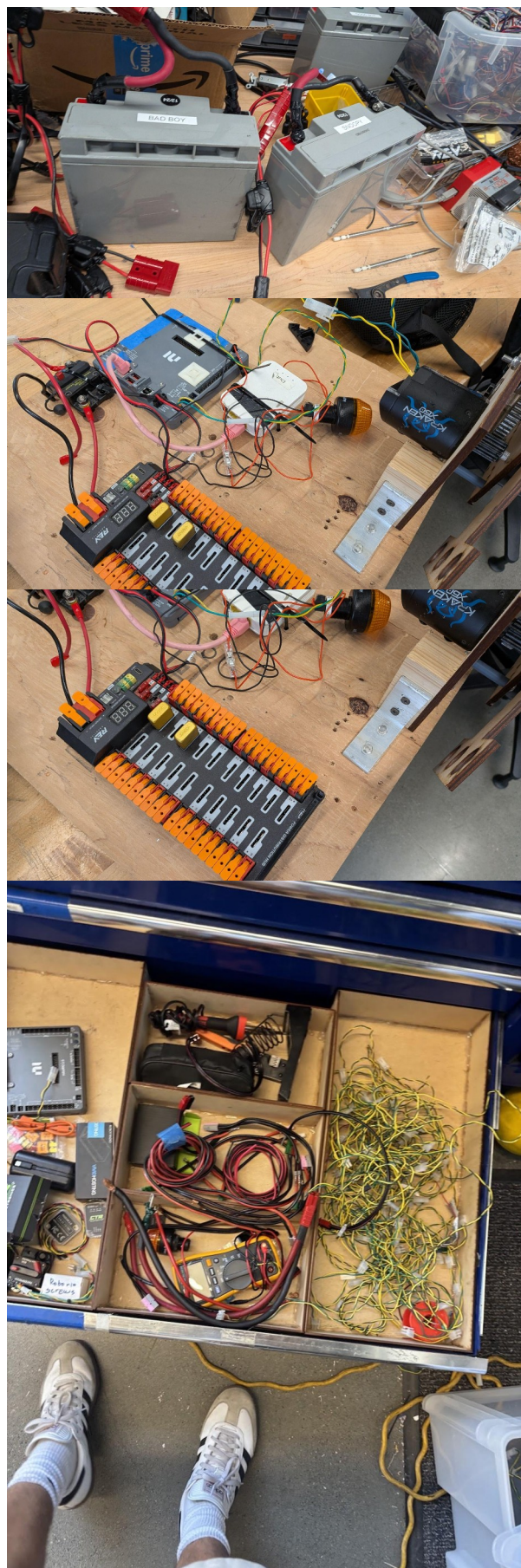
We learned about gauge; and how lower gauge number is a thicker wire than can carry more current, so the battery leads are the thickest wires on the robot. Motors used a more medium gauge and signal is very thin. A thin wire on a high-current channel would simply cause the wire to overheat.

In addition to just power, we covered the parts that carry information through the robot. The radio connects the robot to the driver station and plugs into the RoboRIO over ethernet. The CAN wire is the yellow and green "daisy chains", twisted so the electrical noise cancels out. Since it is a chain, one bad connection can "propagate" down the chain, affecting multiple devices (this ended up being quite important in a future activity...)

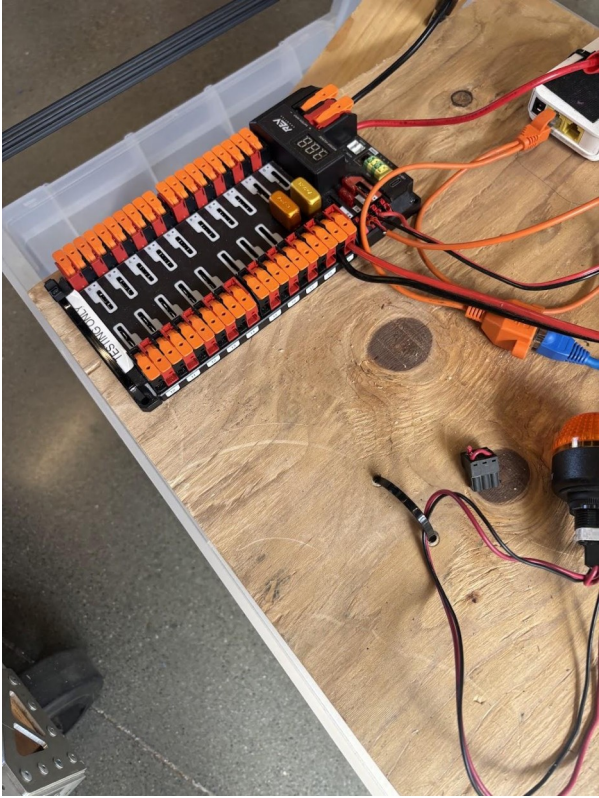
Next, I will quickly cover stripping and crimping:

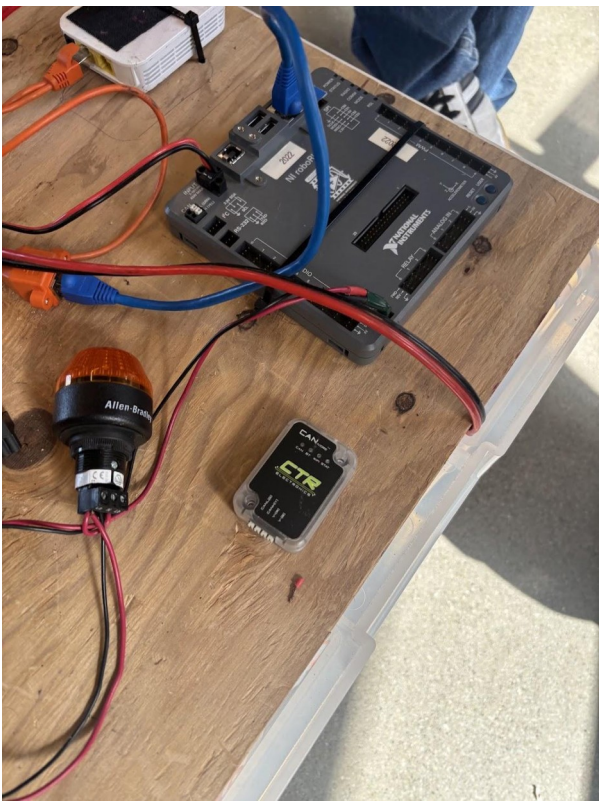
I quickly realized that stripping is very important for crimping: for instance, having too small of a cut leaves a weak wire, and a too large one simply isn't stripped cleanly.

For the crimp itself, the terminal had two sets of wings. One grips the bare copper and the other grips the insulation. One significant challenge I faced is that I would try and put either both on the insulation or both on the copper wire, which didn't work.

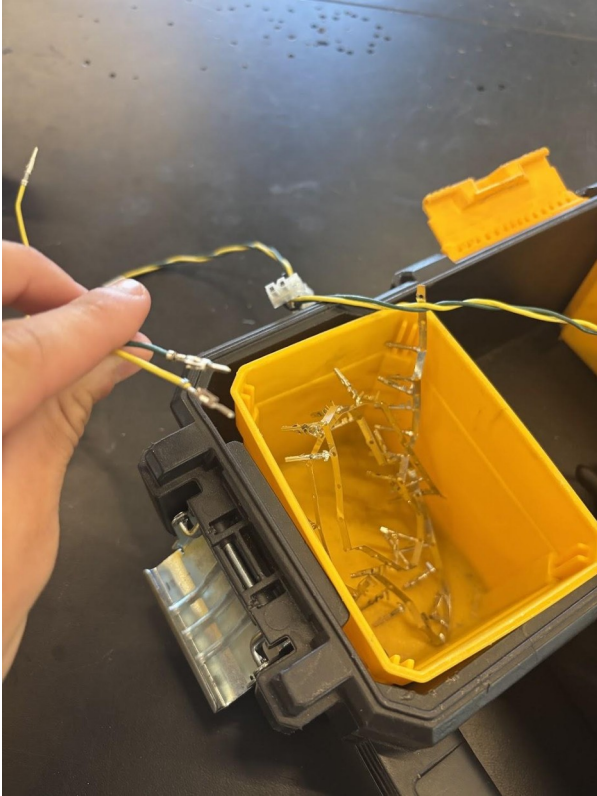
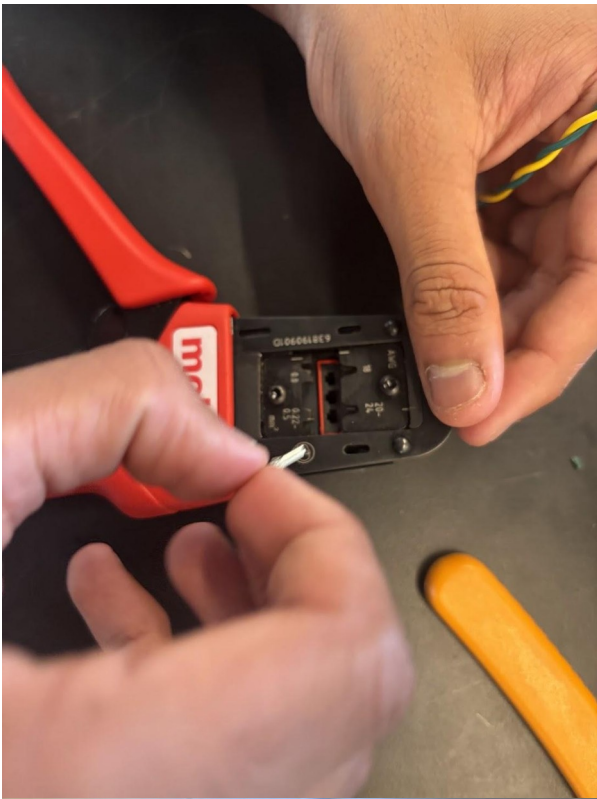


There isn't a lot to speak to regarding individual contribution- it was all "individual" work for today. There was a challenge of simply having a lot to learn.



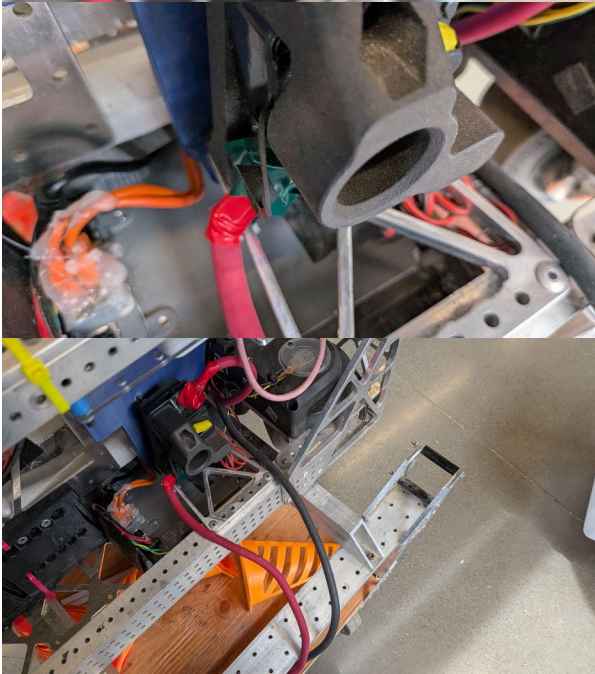








Task 3, learning about the robot and debugging, 1 period.



The final thing we did was a “debugging game” of sorts. Ishika had deliberately tampered with the robot, introducing faults, entirely in the electrical selection (it is the electrical rotation, after all). Our task was to find and fix all the errors.

We immediately noticed the flashing lights on the motors. However, in a CAN chain, one issue can cascade into many others; simply plugging in a CAN wire was part of it. That being said, we still had to analyze the whole robot, and we ended up finding another specific issue; the CAN wire was not properly attached in another location.

Once we cleaned those up, the robot was working again!

One difficult part/challenge to me specifically was that compared to a test board, the robot is incredibly cramped. This is made worse by the fact that the robot has a multitude of very technical terms. Without my lead being here, I found it difficult to remember them.

My key lesson from electrical engineering is that many faults are simply electrical/physical, and having a “plan” to tackle these are much better than just guessing and checking on the test.





